

Bargaining

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Introduction

- Bargaining is the process by which economic agents agree on the terms of a deal.
 - Voluntary exchange occurs when there is a gain from trade.
 - Call this gain a surplus.
 - How is the surplus is divided?
- Bargaining is a basic economic activity

Unstructured Bargaining

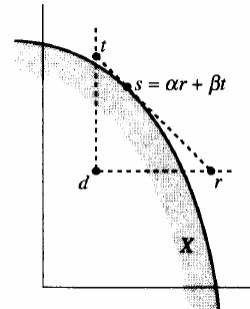
- The details of how the bargaining proceeds are left up to the players
- Camerer suggests it is a better model of naturally occurring bargaining
- Key solution concept is Nash bargaining solution
 - Focuses exclusively on how the surplus is divided
 - Strategic issues are assumed away with a quick reference to communication and contracts
 - This approach is called cooperative game theory

Nash Bargaining Problem

1. Nash Bargaining Problems: a pair (X, d)
 - a. X is convex.
 - b. X is closed and bounded above.
 - c. Free disposal is allowed.
2. The set of all bargaining problems (X, d) satisfying a, b, c, will be denoted B .
3. A *bargaining solution* is a function $F: B \rightarrow \mathbb{R}^2$ with the property that $F(X, d)$ is in the set X .

Generalized Nash Bargaining Solution

- Define $G(X, d)$ to be s in the figure,
 - where α and β are interpreted as the player's bargaining powers;
 - and satisfy $\alpha \geq 0$, $\beta \geq 0$, and $\alpha + \beta = 1$
- The line through r, s, t is a supporting line of X at s , that is, it is a tangent line.
- The function $G: B - \mathbb{R}^2$ defined in this way is called the generalized Nash bargaining solution.
- The regular Nash bargaining solution sets $\alpha = \beta = 1/2$



What properties would be satisfied by a rational procedure for settling bargaining problems?

- P1: Solution is independent of utility scales
 - Rawlsian criterion, maximize the welfare of the worst off individual, is an example of a criterion that is not independent of utility scales.

	L	R
T	2, 2	0, 0
B	0, 0	4, 1

Game G

	l	r
t	1, 4	0, 0
b	0, 0	2, 2

Game G*

- In game G, the Rawlsian criterion selects (T,L).
- In game G*, the Rawlsian criterion selects (b,r).
- But game G* is derived from game G by multiplying the row player payoffs by 2 and dividing the column player payoffs by 2.

Nash's Axioms continued

- P2 (Efficiency): Solution is feasible, individually rational, and efficient

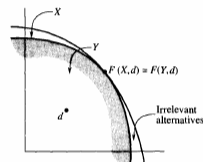


Figure 5.13 Independence of Irrelevant Alternatives.

Axiom 5.3 If $d \in Y \subseteq X$, then

$$F(X, d) \in Y \Rightarrow F(Y, d) = F(X, d).$$

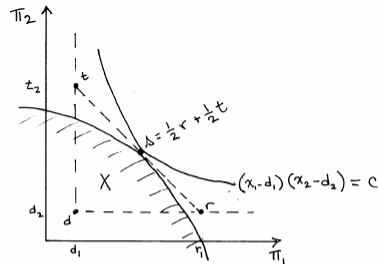
- P3 (Independence of irrelevant alternatives): If players sometimes agree on s when t is feasible, then they never agree on t when s is feasible.

Nash's Axioms continued

The first three axioms give the generalized bargaining solution, but Nash thought there was a fourth property that a rational bargaining solution ought to include.

- P4 (Symmetry): in symmetric situations players are treated symmetrically, that is, they are assumed to have the same bargaining ability.

Nash Bargaining Solution



Theorem: If $N: B \rightarrow \mathbb{R}^2$ satisfies P1 to P4 then the solution corresponds to the (x_1, x_2) that maximizes the Nash Product, that is,

$$\max_{x \in X} (x_1 - d_1)(x_2 - d_2)$$

such that

$$x \geq d.$$

Nydegger and Owen (1975)

- Paired participants bargained over the division of a known prize
- Pairs seated face to face at tables
- Single bargaining decision
- Three treatments
 - Baseline treatment bargain over a dollar
 - All ten pairs divided the dollar equally (50,50), which is the Nash solution in this case.
 - Two asymmetric treatments

Nydegger and Owen (1975)

- Treatment Two: to test independence-of-irrelevant alternatives axiom
- Again, bargained over the division of a dollar, but one of the players was constrained to earn no more than 60 cents.
- By IIA, the (50,50) division remains the Nash Solution
- All 10 pairs divided the dollar equally in the pairings with truncated payoff possibilities.

Nydegger and Owen (1975)

- Treatment Three: test invariance-to-linear-transformations-of-utility axiom.
- Again ten pairs engaged in face-to-face bargaining
- Bargained over the division of 60 poker chips
- Common knowledge that each chip was worth one cent to player 2 and two cents to player 1.
- The Nash solution divides the chips equally (30 chips each), but this gives unequal money payoffs player 1 receives 60 cents and player 2 receives 30 cents.
- All ten pairs split the money equally (40 cents each) rather than the chips equally.

Roth and Malouf (1979)

- Nydegger and Owen (1975) result rejects the joint hypothesis of utility-invariance axiom and risk neutral own-money motivated preferences.
- Roth and Malouf (1979) used a binary lottery technique to induce risk-neutrality.
 - Bargain over 100 lottery tickets
 - Each player has a fixed cash prize
 - The number of tickets determines their chance of winning the cash prize
 - For example, 72 tickets gives 0.72 chance of winning the prize and 0.28 chance of winning nothing

Roth and Malouf (1979)

- Treatments
 - Prizes
 - Equal money prizes (\$1,\$1)
 - Unequal money prizes (\$1.25, \$3.75)
 - Information
 - Full information – both subjects know the prizes
 - Partial information – subjects only know their own prizes
- Under all conditions the Nash Bargaining solution predicts that the 100 chips will be divided evenly.

Roth and Malouf (1979)

Table 4.1. Results in binary lottery bargaining games

Information condition	Money prizes	Number of tickets for player 2							Fraction of disagreement
		20	25	30	35	40	45	50	
Full	(\$1,\$1)	0	0	1	0	1	0	20	0.00
	(\$1.25,\$3.75)	1	6	3	2	2	1	4	0.14
Partial	(\$1,\$1)	0	0	0	0	0	1	14	0.06
	(\$1.25,\$3.75)	0	0	0	0	0	3	13	0.00

Source: Roth and Malouf (1979).

Alternating-Offer Bargaining

- Formalize the bargaining process in order to use “noncooperative” game theory to make predictions.
- Players often alternate making offers in naturally occurring bargaining.
- Delay is costly because
 - The amount being bargained over loses value
 - There is a fixed cost to delay

Two Stage Bargaining Game

- Restrict $x \in [0,1]$
- Assume discrete time.

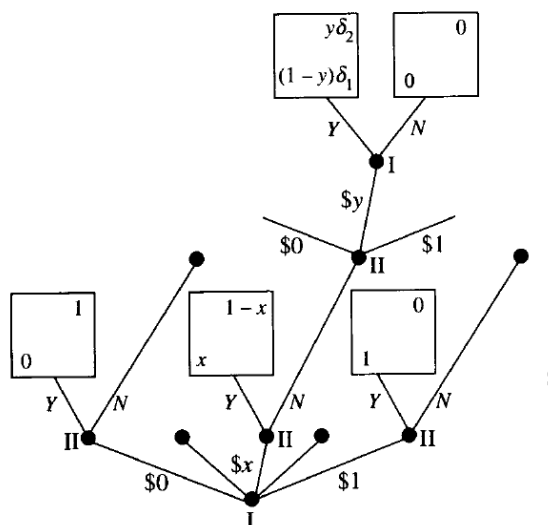
$$u_i(x) = x$$

- Discount factor:

$$u_i(x, t) = x\delta_i^t$$

$$0 < \delta_i < 1$$

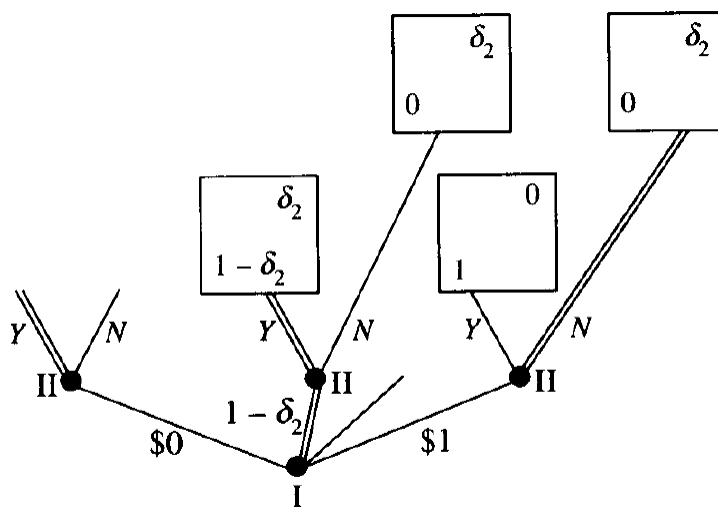
Two Stage Bargaining Game



Two Stage Bargaining Game

- Solve using backwards induction.
- Final stage is an ultimatum game, and we know the unique SPE has $y = 1$.
- The outcome of the second stage is $(0, \delta_2)$.

Pruned Two Stage Bargaining Game



Two Stage Solution

- Player 1 must leave Player 2 indifferent between accepting proposal and making the final counter proposal.
- Unique SPE for the two stage game is $(1-\delta_2, \delta_2)$.
- Player 1's share depends on how impatient player 2 is.
- If δ_2 is close to one, Player 2 gets almost all of the surplus.
- Changing the rules "slightly" gives almost the opposite prediction.

Binmore, Shaked, and Sutton (1985)

- Two period game
- First period, player 1 offers a division of 100 British pence to player 2
- If player 2 rejects it, then the "pie" being divided shrinks to 25p (that is, $\delta = 0.25$) and player 2 makes a counter-offer.
- The SPE are centered around a initial demand of 75p, leaving 25p for player 2.

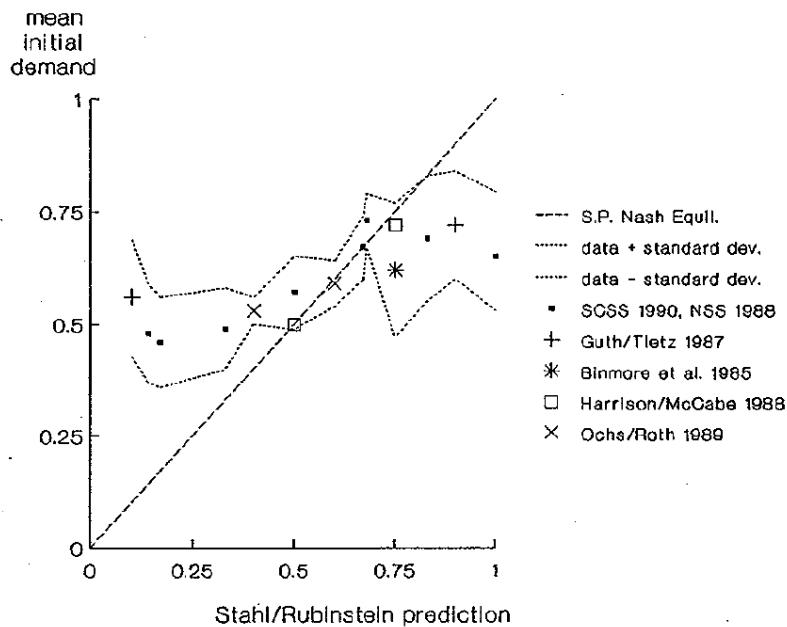


Figure 5.6 Mean Initial Demands in Alternating-Offer Bargaining Games with Two Rounds (Source: Spiegel et al., 1990, table 6)

Ochs and Roth (1989)

- The number of rounds was two or three
- The discount factor was 0.4 or 0.6
- And the discount factor could be different for different players
- Offers are out of \$30.

Ochs and Roth (1989)

Table 4.5. *Predictions and offers in alternating-offer bargaining*

Cell	Number of rounds (T)	Discount factors		Perfect equilibrium	Average opening offer		Rate of rejection
		δ_1	δ_2		Round 1	Round 10	
1	2	0.4	0.4	12.00	13.19	12.03	0.10
2	2	0.6	0.4	12.00	14.73	14.34	0.15
3	2	0.6	0.6	18.00	13.88	14.70	0.13
4	2	0.4	0.6	18.00	14.67	13.57	0.20
5	3	0.4	0.4	7.20	13.02	12.81	0.12
6	3	0.6	0.4	4.80	14.04	13.17	0.14
7	3	0.6	0.6	7.20	13.93	13.70	0.15
8	3	0.4	0.6	10.50	13.90	14.23	0.29

Source: Ochs and Roth (1989).

Ochs and Roth (1989)

- Counteroffers that follow rejections have two interesting properties
 - Second period counteroffers are rejected more often (40 percent) than opening offers are and third-period rejections are even more common (54 percent)
 - Most (81 percent) counteroffers are “disadvantageous”
 - A player who rejects an offer then makes a counteroffer that leaves herself less than she was offered.
 - Disadvantageous counteroffers are common in other studies

Limited Computation

- The main question about alternating-offer bargaining results is what combination of stable social preferences and failures of backward induction explains what happens.
- Camerer et al. (1994) and Johnson et al. (2002)
 - Subjects play against a computer that was programmed to maximize its own earnings.
 - Subjects should play SPE if they can compute it.
 - The second design feature was that the “pie sizes” in each of three rounds of bargaining were hidden in boxes on a computer screen.
 - Subjects should search for information in a particular way if they are using backwards induction.

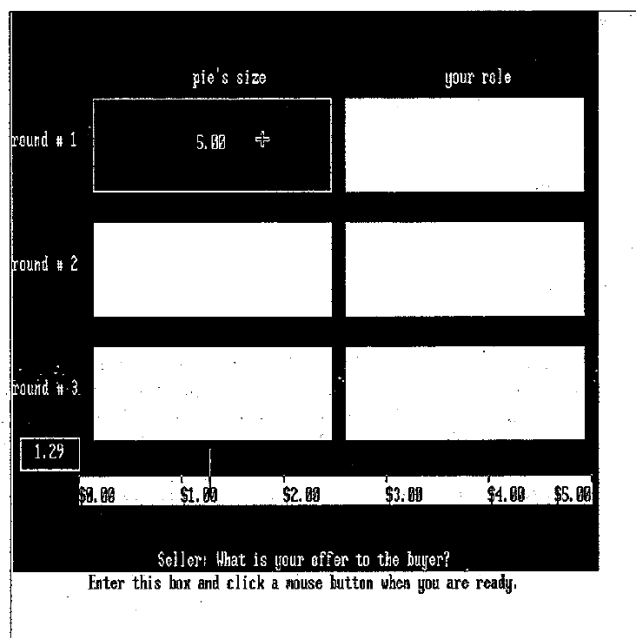


Figure 4.1. The information display in alternating-offer mouselab experiments. Source: Johnson et al. (2002), p. 22, Figure 1; reproduced from Journal of Economic Theory with

Johnson et al (2002)

- Subjects bargained over three periods in which the pie sizes were around \$5.00, \$2.50, and \$1.25.
- To prevent memorization, the pie sizes were perturbed in each period such that the subgame perfect prediction was fixed at \$1.25
- The bargaining results replicate earlier findings:
 - Most offers were between \$2.50 and \$2.00, averaging \$2.11
 - Offers less than \$1.80 were rejected about half the time.
 - The overall rejection rate was 12 percent.

What did subjects play attention to?

Table 4.6. Average lookups, gaze times, and transitions by player 1s

Round	Pie size (\$)	Number of lookups	Fraction of periods with zero lookups	Gaze time	Number of transitions from row round to column round		
					1	2	3
1	5.00	4.38	0.00	12.91	—	2.55	0.65
2	2.50	3.80	0.19	6.67	2.10	—	1.24
3	1.25	2.12	0.10	1.24	0.50	0.88	—

Source: Johnson et al. (2002).

Johnson et al (2002)

- A key fact is that players did not open the second and third round boxes at all on 19 percent and 10 percent of the trials, respectively.
- Rejects a strong form of backward induction hypothesis.
- Analysis of the lookup patterns also suggests they are not backward inducting.
- The basic look up direction is from round 1 to 2 to 3, not the reverse.

Computer opponents and instruction

- Against computer opponents know to play the SPE the average offer was \$1.84, still well away from the SPE prediction of \$1.25.
- After brief instruction in backward induction, subjects lookup pattern and gaze time were quite different: They looked mostly at the second and third round boxes and made more transitions between those boxes.
- The average offer in those periods was \$1.22.

Conclusion

- Modeling social preferences is insufficient to explain these results
- Note well that the attentional statistics and results from computerized opponents also reject the new generation of social preference theories that assume equilibrium reasoning.
- After training in backward induction, they do make equilibrium offers to computerized opponents so they are cognitively capable of backward induction in three stage alternating offer bargaining games, they just don't do it.